Oriental Institute of Science and Technology
Bhopal

Department of Artificial Intelligence & Machine Learning

SuhanaGhar AI

Climate-Responsive Smart Housing System

Progress Seminar 1

Session 2025-2026

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SOFTWARE DEVELOPMENT LIFE CYCLE

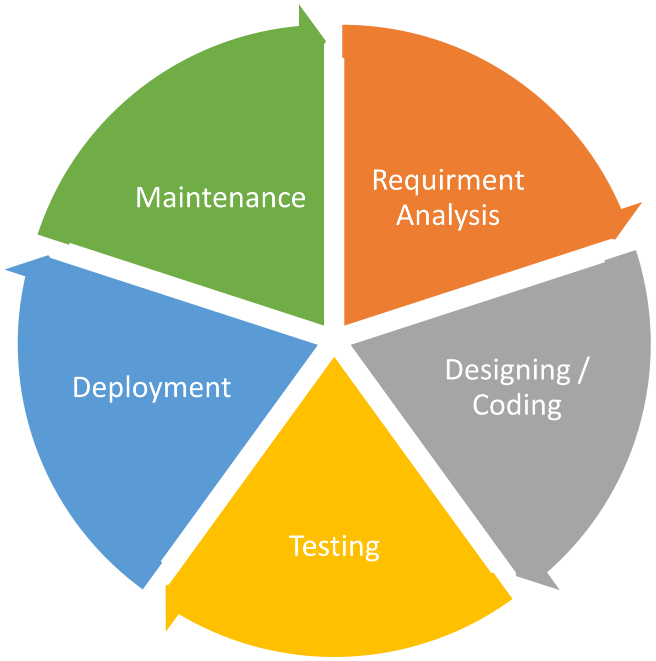
Requirement Analysis : Gathering needs: energy prediction, design recommendations, climate detection

Designing/Coding : Streamlit frontend + Python backend + Random Forest ML model

Testing : Model accuracy (R^2), UI response time, prediction quality

Deployment : Local/cloud hosting for public access

Maintenance : Updates, model retraining, new features



Introduction & Problem Statement

WHAT IS SUHANAGHAR AI?

A Machine Learning web app that predicts heating and cooling loads and provides climate responsive design recommendations

PROBLEM STATEMENT

Traditional Indian homes ignore:

- Local climate conditions
- Local climate conditions
- Building orientation
- Building orientation

RESULT: High electricity bills, poor comfort, High carbon footprint

Project Modules

MODULE

Function :

🔵 AI Analysis : Predicts heating/cooling load using 8 parameters

🏠 Design Advisor : Climate recommendations based on city

📘 Feature Guide : Explains impact of each parameter

Input Parameter :

Relative Compactness, Surface Area, Wall Area, Roof Area, Overall height, Orientation, Glazing Area, Glazing Distribution

Flowchart

User Input

- Simple Mode
- AI Analysis
- Design Adviser

- Quick Score
- ML Model Predicts
- Wheater API Fetches Data

Display Result

- Charts + Score + Recommendations

Technology Stack

Category	Technology
Frontend	Streamlit
Backend	Python
Machine Learning	Sckit – Learn (Random Forest)
Data Handling	Pandas, Numpy
Visualization	Matplotlib, Plotly
API	Requessts (Weather)

Results & Achievements

Metric	Achievement
Heating Load Accuracy	$R^2 = 0.924$
Cooling Load Accuracy	$R^2 = 0.896$
Prediction Time	< 1 second
Parameter Supported	8 features
Cities Supported	50+ Indian Cities

SLIDE 8: Conclusion & References

SuhanaGhar AI successfully demonstrates Machine Learning for sustainable architecture, making energy efficiency accessible to everyone

Future Scope : Google Maps Integration, 3D visualization Cost estimation module, Solar panel recommendation

References :

- Scikit-learn Documentation
- UCI Energy Efficiency Dataset
- Tsanas & Xifara (2012) – Energy and Buildings

THANK YOU

SuhanaGhar AI – Your home, made smarter.